

NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

Collision avoidance sensor

Toposens Echo One is the world's first industrial-grade 3D ultrasonic collision avoidance sensor. It employs three microphones to detect sound



waves coming from different angles. Echo One is capable

of detecting multiple difficult-to-detect obstacles reliably and in real time, regardless of environmental conditions. The sensor sends the data to Toposens processing unit, which uses sound based triangulation combined with sophisticated noise-filtering software to detect any obstacle. The sensor addresses the shortcomings of optical sensors, such as lidars, which can fail to detect floor-based objects like fork-lift forks on the floor in unfavourable lighting conditions or on transparent or mirrored object surfaces.

Toposens

<https://toposens.com>

Highly integrated BLE SoC

SmartBond DA1470x from Renesas is a Bluetooth Low Energy (BLE) system-on-chip (SoC). It is the first BLE solution in the market to feature



a 2D graphics processor, voice activity detector, and power man-

agement unit in a single chip. The SoC is powered by an ARM Cortex M33 as the main processor and an ARM Cortex-M0+ as a sensor node controller. The small form factor and high level of integration results in achieving lower costs in bill of materials (BoM), thus enabling designers to make compact, cost-effective system solutions.

The SoC features ultra-low-power, always-on audio processing. The product can be used in a wide variety of applications, including wearables and security systems.

Renesas

<https://www.renesas.com/in/en>

ESD protection devices

PESD2V8Y1BSF and PESD4V0Y1BCSF from Nexperia are electrostatic discharge (ESD) protection devices optimised for use with re-timers



and re-drivers on high-speed data lines. Re-timers and re-drivers are commonly used in the

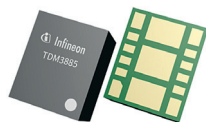
design of high-speed USB4 interfaces. The PESD2V8Y1BSF is designed to protect USB4 (Thunderbolt) interfaces, whereas PESD4V0Y1BCSF can protect both the USB4 as well as the HDMI 2.1 interface. The component employs Nexperia's TrEOS technology, which combines low clamping with low capacitance and high robustness. Both the devices offer a backward compatibility and can also be used to protect PCIe or DisplayPort interfaces.

Nexperia

<https://www.nexperia.com>

DC-DC converter modules

TDM3883 and TDM3885 IPOL from Infineon are fully integrated single-



output step-down DC-DC converter modules. The high-efficiency

buck converters are capable of outputting continuous 3A/4A load capability and line regulation over a wide input supply range (4.5V to 14V). Both

modules improve the battery life of the application device due to a low supply current (10µA) at shutdown and tight reference voltage VREF tolerance (0.5V ± 1.0%) and low ripple. The modules come with thermally compensated internal over-current protection (OCP). The buck regulator modules are specially designed for space and thermally constrained applications, telecom and datacom applications, servers, and network storage.

Infineon

<https://www.infineon.com>

Mode-programmable boost converter

AP72250 from Diodes Incorporated is a high-efficiency, mode-programmable, synchronous boost converter with multiple operating modes. Different



modes allow adjusting power and efficiency based on the requirements of the operation, making

the module capable of achieving a switching frequency of up to 900kHz. The converter is designed for low voltage products that need to achieve compact size in step-up conversion applications, such as battery-powered devices, USB power supply units, power banks, supercapacitor charging equipment, and metering systems.

Diodes Incorporated

<https://www.diodes.com>

AR/VR sensor for smartphones

STM VL53L8 is a sensor for smartphone cameras that can enhance the augmented reality (AR) and vir-



tual reality (VR) capabilities of the device. The module features a

94nm VCSEL light source, a low-power 32-bit MCU core and an SoC sensor